Recursive Self-Evolving Agents via Held-Out Selection

Michael Nguyen · Quoc Nguyen · Paul Vuong | Monash University Malaysia | arXiv 2606.28374v1

HarnessRSI 67 ★★

evidence: retained

Paper

1 What changes, what is kept

Retained state = natural-language preamble
STRATEGY · SKILLS · PLAYBOOK
Frozen LLM + unchanged ReAct/tool interface

1 Roll out on evolve tasks
Collect successes, failures and verified outcomes.
Start with empty working and best states.

2 Rewrite all three layers
Same frozen LLM reads balanced trajectories.
It can rewrite or delete any layer.

3 Reuse disjoint validation tasks
Candidate score ≥ current → accept working state.
Otherwise keep current state. Ties may advance.

4 Freeze only a strict new best
Candidate score > historical best → replace best.
No strict gain over empty → return vanilla ReAct.

5 Run frozen best on separate test
RSEA: single pass. RSEA+retry: within-task retry.
Validation is the selection set, NOT the test.

Algorithm 1, p5. Validation winners are selected by construction. The guarantee is on validation, not every unseen test draw (§10).

2 ALFWorld: a paired-test gain

Qwen2.5-7B-Instruct · 134 tasks × 5 seeds
Success %: mean ± 95% CI (Table 2)



Vs ReAct: paired p=0.015. DC has the higher single-pass mean (p=0.45 vs RSEA); "strongest" wording unresolved.

Retry vs Reflexion / ACE / GEPA: p=0.09 / 0.13 / 0.052, not significant.

4 Where gains stop

ALFWorld families (ReAct → RSEA %):
examine 6.7 → 26.7; pick_two 36.5 → 43.5;
pick_and_place 82.5 → 91.7.
Clean regresses: 80.0 → 73.5 (Table 3).

Test (Table 4)	ReAct	RSEA	Best base
GAIA % · n=30	16.7	13.3	16.7
τ-bench % · n=60	41.7	40.0	51.7
WebShop · n=100	0.429	0.437	0.460

Qwen3-30B-A3B-Instruct; one split seed, no CIs. Best: ReAct / AWM / AWM. GAIA: two methods only; ≤ one-task gap flips across reruns.

WebShop keeps empty state; DC: 0.136. Not an isolated gate ablation.

3 Controls, not just the headline

Layers overlap · ALFWorld (Table 5)

Injected state	Success %
Empty / strategy	64.2 / 67.3
Skills / playbook	69.8 / 69.8
Strategy+skills / full	67.9 / 68.5

54 tasks × 3 seeds; means, no per-cell uncertainty. Full state does not beat the best single layer.

Selection ablation · Table 6

Variant	In-sample	Test %
Empty ReAct	—	63.6
Select on evolve	100.0	66.7
Strict val gate	—	67.3

The 33-point gap is train→test, NOT a gate benefit. No CI; exact task/seed count not stated.

- **Held out:** evolve / val / test disjoint. Val gates adoption repeatedly. Tool-use: shuffled test-pool slices, not distribution shift.
- **Baselines:** GEPA/ACE have native val selection. No strict-gate control for every baseline; retry scaffold shared.
- **Budget:** equal rollouts asserted, not total compute. τ-bench: 190 meta+selection calls; no per-method cost table.
- **Unreported:** baseline tuning budgets; hyperparameter/default selection split; exact evolve/val sizes and rollout/retry limits.

Takeaway

A retained procedural prior helps on ALFWorld; it does not win everywhere. Separate candidate-writing, selection and test data. A monotone validation curve is not a test-safety guarantee.

Our read · 67 · retained

Reports: held-out test, seeds, ablations.
Not established: total-compute match, artifact transfer, contamination audit, evolved optimizer.